Set-up guide

LED Letters

Turn your printed letters into a glowing sign. Install the software once, then set everything up from your phone. No app and no coding.



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You only do steps 1 to 4 once. After that, the sign remembers everything.

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A word about "glyphs"

A glyph is one printed piece of your sign: a letter, a digit or a symbol. You can use the letters A–Z, the digits 0–9 and the symbols &, @, # and ♥, up to six of them in one sign.

1 What you need

Item	Notes
Your glyphs	Printed and fitted with LED strip (see step 3). Up to six in one sign.
The controller	A small ESP32-C3 board with a USB-C socket and two tiny buttons.
A USB-C cable	One that carries data, not only charging, for installing the software.
A power supply	A good USB phone charger (1 A or more) for up to four glyphs. For five or six glyphs, a stable 5 V supply (see step 8).
A computer	Only for installing the software once. Chrome or Edge browser, Windows, Mac or Linux.
A phone	Any smartphone with Wi-Fi and a web browser. No app needed.

How it works

The controller sends light to the glyphs one after another, like a chain. It also makes its own small Wi-Fi network, called a **hotspot**. You join it with your phone and a page opens where you set everything up. Your home Wi-Fi is not needed, but you can connect the sign to it later if you like.

2 Install the software

You do this once, before connecting the glyphs. It takes about two minutes and nothing needs to be installed on your computer.

1. **Download the software.** Go to **github.com/maciejfic/led-letters/releases** and download the file `led-letters-esp32c3-full.bin` from the newest version.
2. **Plug the controller into your computer** with the USB-C cable.
3. **Open the installer page** in Chrome or Edge: **espressif.github.io/esptool-js**. This is the official tool from the maker of the chip.
4. **Click "Connect"** and choose the controller from the list. It is usually called "USB JTAG/serial debug unit". Click "Connect" in the small window.
5. **Choose the file.** In the "Flash Address" box type `0x0`, then click "Choose File" and select the file you downloaded.
6. **Click "Program"** and wait until it says it has finished.
7. **Unplug the controller.** The software is installed.

The controller is not in the list?

Try another USB-C cable: some only charge. Still nothing? Hold the button marked **BOOT**, press and release **RESET**, let go of BOOT and click "Connect" again.

Updating later

To install a newer version and keep your settings, use the file `led-letters-esp32c3-update.bin` with Flash Address `0x10000`. The "full" file starts fresh and forgets your glyphs and Wi-Fi.

Prefer the Arduino IDE? The GitHub page has instructions for that too.

3 Connect the glyphs

Follow the soldering guide

The **LED Letters soldering guide** (LED_Letters_Soldering_Guide.pdf, published with the 3D files on MakerWorld) shows, for every glyph, how to fit the LED strip pieces, where each wire goes and how many LEDs each piece has.

Good to know before you start

- Each glyph has two short cables with **3-pin plugs**: **IN** and **OUT**. The glyphs simply plug together: the OUT of one glyph into the IN of the next.
- The glyphs form a **chain**. The controller's cable goes into the IN of the **first glyph**, that one passes the signal on to the second, and so on.
- The order of the chain is the order you will enter in the app, and normally the order you read the sign, from left to right.
- Inside every glyph the LED pieces are wired in a fixed order. Follow the guide's order: the app knows it, so effects like Rainbow sweep and Fire move across your sign the right way.

LED counts are not yet verified

The LED count of each glyph in the guide and in the app is the planned number, and has not been checked on every built glyph yet. Count the LEDs in yours. If the number is different, that's fine: you enter the real number in the app during set-up (step 4).

4 Set up from your phone

Switch on

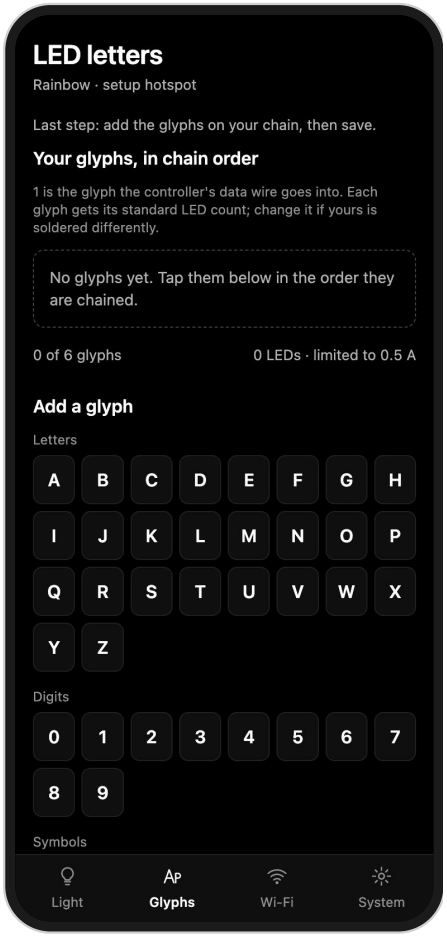
Plug the controller into its power supply. Before you set it up, the LEDs show a moving rainbow, so you can see right away that everything is connected.

Join the hotspot

1. On your phone, open the **Wi-Fi settings**.
2. Choose the network **LED-Letters-Setup**. It has no password.
3. The set-up page opens by itself. If it doesn't, open your browser and go to **http://192.168.4.1**

> **The hotspot is there for 10 minutes**
>
> After switching on, the hotspot stays available for 10 minutes, and for as long as your phone is connected to it. Missed it? Unplug the sign and plug it back in.

Your phone may say "no internet" on this network. That's normal: stay connected.

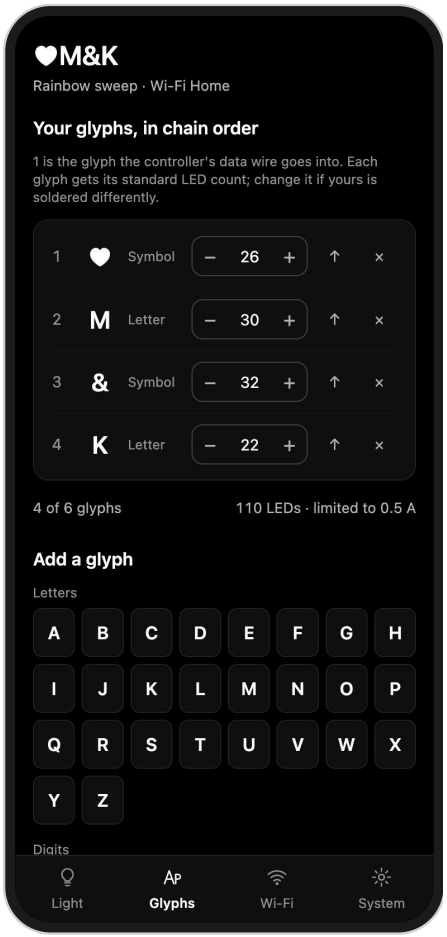


The set-up page on first start

Add your glyphs

1. **Tap your glyphs in chain order**: first the one the controller's wire goes into, then the next one, and so on. For a sign that reads ♥M&K, tap ♥, M, & and K. Letters, digits and symbols are in separate groups, so you can't mix up the letter O and the digit 0.
2. **Check the numbers.** Each glyph shows its standard number of LEDs. If yours has a different number, change it with `−` and `+`.
3. **Fix the order** if needed with `↑`, or remove a glyph with `×`.
4. **Tap "Save and restart".** The sign restarts in a few seconds with your glyphs.

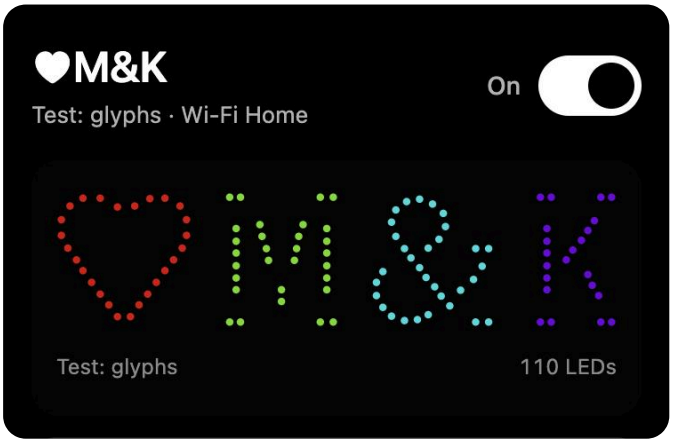
> **More than four glyphs?**
> With five or six glyphs the app shows a reminder: they need a very stable power supply (see step 8). Six is the maximum, because more would overload the power the controller can carry.



Four glyphs added

Check that it's right

Tap **"Check glyphs on the sign"** under the Save button. Every glyph lights up in its own colour, and the preview at the top of the app shows what you should see:



- **Each glyph is one colour, from its first to its last LED:** perfect, you're done.
- **A colour runs into the next glyph:** the first glyph has more LEDs than you entered. Increase its number.
- **The last LEDs of the sign stay dark:** a glyph has fewer LEDs than you entered. Find it and decrease its number.

Then choose any effect on the Light page to leave the check.

5 Connect to your Wi-Fi optional

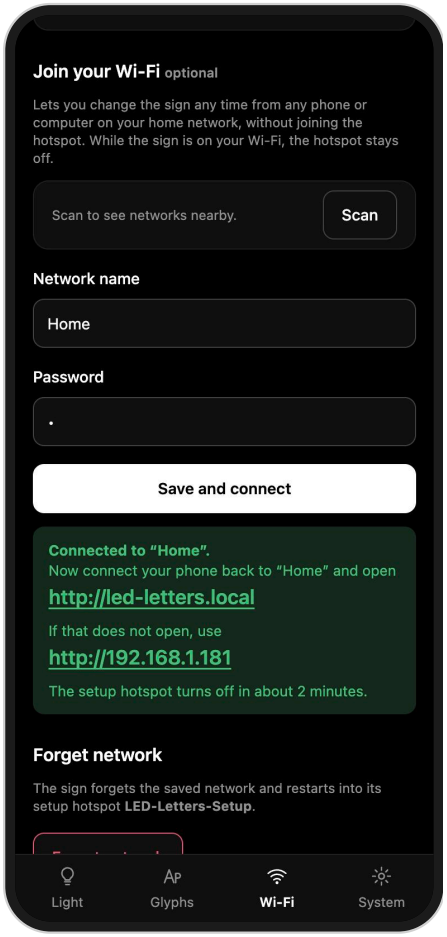
Without Wi-Fi the sign works just the same and keeps all its settings. You can always change things over the hotspot, within 10 minutes of switching it on. Connect it to your home Wi-Fi if you want to change effects at any time, from any phone or computer at home.

1. While connected to the hotspot, open the **Wi-Fi** page in the app (bottom bar).
2. Tap **Scan** and choose your home network, or type its name.
3. Enter your Wi-Fi password and tap **Save and connect**.
4. Wait up to 20 seconds. The page then shows **"Connected"** and the sign's address on your network.
5. Connect your phone back to your home Wi-Fi and open the address. **Bookmark it** or add it to your home screen.

The address

On most phones and computers: **http://led-letters.local**
If that does not open (some Android phones), use the number shown, for example **http://192.168.1.181**. You can always find both on the app's Wi-Fi page.

Once the sign is on your Wi-Fi, the setup hotspot stays off. To go back to the hotspot, use "Forget network" on the Wi-Fi page, or hold the button for 6 seconds (step 7).



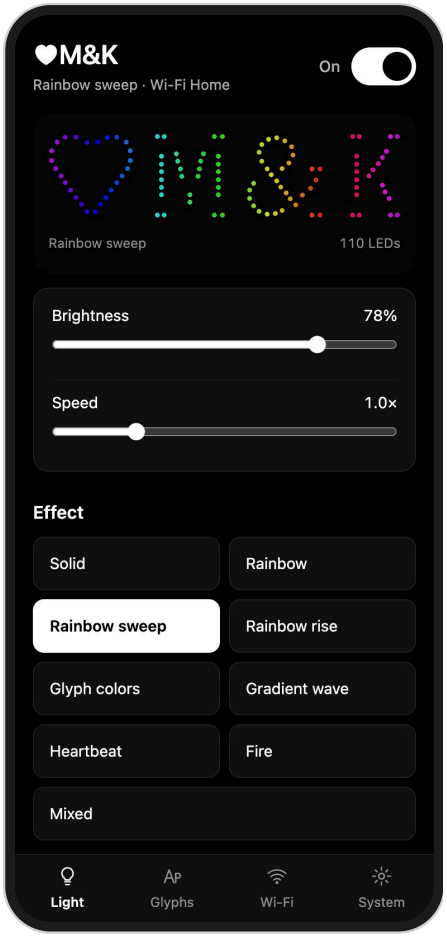
Connected, with the sign's address

6 Everyday use

The **Light** page is where you'll spend most of your time. The picture at the top is a live preview of your sign: it shows the effect you choose before you even look up.

- **On / Off**: the switch at the top right.
- **Brightness** and **Speed**: the two sliders.
- **Effect**: tap one of the buttons.
- **Colours**: some effects use your two colours (below the effects).
- **Change effect automatically**: let the sign switch effects every few seconds or minutes.

Everything is saved on the sign straight away and kept when it's switched off.



The Light page

The effects

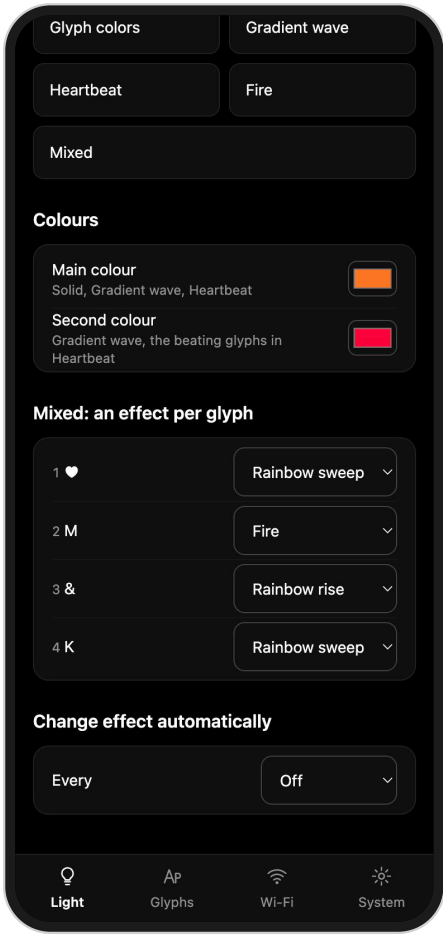
Effect	What it does
Solid	The whole sign in your main colour.
Rainbow	Rainbow colours running along the sign.
Rainbow sweep	A rainbow moving from left to right across the sign.
Rainbow rise	A rainbow rising from bottom to top.
Glyph colors	Every glyph in its own colour, slowly changing.
Gradient wave	Your two colours flowing across the sign.
Heartbeat	♥ and & beat in your second colour; the other glyphs glow softly in your main colour.
Fire	Flickering flames, hotter at the bottom.
Mixed	A different effect on every glyph, chosen below the colours.

Mixed: an effect per glyph

Choose an effect for each glyph, for example fire on the M and a rainbow on the heart. Choosing one switches the sign to Mixed.

Colours

Tap a colour square to pick a colour. The main colour is used by Solid, Gradient wave and Heartbeat; the second colour by Gradient wave and the beating glyphs in Heartbeat.



Colours and Mixed

7 The button

The controller's **BOOT** button works the sign without a phone.

Press	Does
Short press	Next effect. If the sign is off, it turns it on.
Hold about 1 second	Turns the sign on or off.
Hold 6 seconds	The sign blinks blue, forgets your Wi-Fi and opens the setup hotspot again. Your glyphs and effects are kept.

The other button, RESET, just restarts the controller.

8 Power and safety

- **Up to four glyphs:** a good USB phone charger (1 A or more) is enough.
- **Five or six glyphs:** use a good quality, stable 5 V power supply. If LEDs flicker or change colour, the supply is not stable enough or the wires are too thin.
- **Six glyphs is the maximum.** More would overload the controller and its wiring.

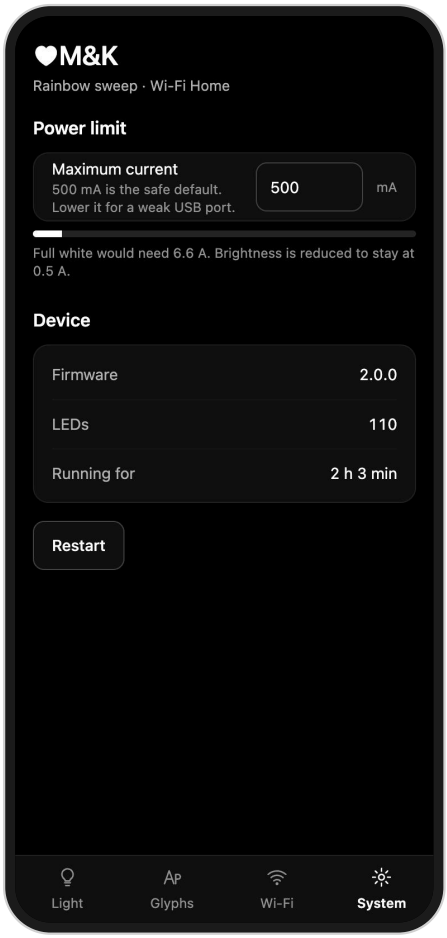
The power limit

To protect the controller and the wiring, the sign never uses more than its **power limit**: **500 mA** unless you change it. When an effect would need more, the sign dims itself a little.

Raising the limit

You can raise it on the **System** page, but above 500 mA you risk damaging the controller, the wiring and the power supply. Only do it if all of them are made for the higher current.

Do not cover the glyphs or leave the sign in direct sunlight. Use it indoors only.



The System page

9 If something is not right

Problem	What to do
No LEDs light up	Check the power supply and the cable to the first glyph. Make sure the controller's wire goes into the **input** of the first glyph.
I can't find LED-Letters-Setup	The hotspot is only there for 10 minutes after switching on, and not at all while the sign is on your Wi-Fi. Unplug the sign and plug it back in. If it is on your Wi-Fi, use its address instead, or hold the button for 6 seconds.
The set-up page doesn't open	Stay connected to LED-Letters-Setup and open **http://192.168.4.1** in your browser. Turn off mobile data for a moment if your phone keeps leaving the network.
Wi-Fi: "Could not join"	Check the network name and password (they are case-sensitive). The sign needs a 2.4 GHz network; most home routers have one.
led-letters.local doesn't open	Use the number address shown on the app's Wi-Fi page, or look for "led-letters" in your router's list of devices.
Colours spill into the next glyph	An LED count is wrong. Use "Check glyphs on the sign" (step 4) and correct the numbers.
LEDs flicker or show random colours	Use a better power supply or shorter, thicker wires, and lower the brightness.
The sign is dimmer than expected	It is staying under the power limit (step 8). Lower the brightness a little, or use fewer glyphs per controller.
I want to start completely fresh	Install the software again with the "full" file (step 2). That forgets everything.

More help

Software, updates and technical details: **github.com/maciejfic/led-letters**
3D-printable glyphs: search for **"LED Letters"** on **makerworld.com**